In Silico De Novo Design and Validation of Synthetic RNA Thermoswitches Via Deep Generative Modelling

Msc Data science, University of Aberdeen 2026

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Abstract

44 Bacteria like Escherichia coli (E. coli) are utilised as powerful cellular factories for highly valued foreign chemicals. However, optimising yield demands precise control over the onset of the synthesis. Premature leaky expression initiates a cascade of unwanted mutant E. coli, which destroys volumetric yield. RNA thermoswitches (RNAT) offer a robust yet chemically-free solution. The lock, the stem loop structure in a RNAT, sequesters the ribosome binding site (RBS), repressing translation at 37°C, allowing the E. coli to populate to maximum biomass prior to induction. This lock then melts at 42°C, triggering production of the targeted foreign chemical.

Initially, Random Forest models trained on natural Rfam and background RefSeq sequences were used to predict and classify viable thermoswitches from unseen sequences. This proved inadequate. Classifiers trained on static scalar features simply could not capture the complex dynamics of the multi state thermodynamics of RNA folding. As such, the Evolutionary Versatile Architect 2 (EVA) model was a natural pivot to engineer novel RNAT candidates fine tuned to this precise heat shock window alongside an *in silico* pipeline to evaluate 105 yield-gated EVA candidates against ~~the~~ 45
2,396 controls; opposed to a costly in vitro screening, to extract Hill functions metrics to compare switching steepness and activation thresholds.

Whilst the EVA model successfully generated topologically diverse sequences passing 46
and learning the baseline ~~biological physical architecture~~ at a repressed state, they universally failed the strict dynamic thermal test, rather than executing the targeted sharp, synthetic phase transition at heat-shock threshold, the engineered sequences remained statically occluded at heat shock and denatures at 42°C.

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1.Introduction

1.1 Yield dilemma in Industrial Biomanufacturing

Modern biomanufacturing is deeply rooted in programmable cellular factories. As 1 such, microbes such as E. coli are fundamental in the production of the desired foreign chemical ranging from pharmaceuticals and industrial enzymes. Living bacteria are not inherently machinery. This introduces a conflict of resources between the bacterium's own goal to survive and multiply and the engineer's goal of chemical production. Engineers follow a strict two-phase bioproduction (Neupert *et al*., 2008): biomass accumulation at 37°C followed by a form of chemical trigger to onset synthesis. Historically, these triggers were chemical inducers which were undermined by their natural flaw of gradual basal leakiness (Neupert *et al*., 2008). This premature expression parasitically eats away at the E. coli's growth. When that burden is stripped away, these mutant E. coli gain a genetic advantage and outcompete the desired yield 2 batch. While protein-based thermosensors such as RheA offer an alternative thermal trigger (Servant *et al*., 2000), engineered RNATs provide a more direct, computationally 3 tractable solution by functioning as a temperature-sensitive lock within the 5' UTR (Kortmann and Narberhaus, 2012) where at 37°C, the RNA folds into a hairpin structure that hides the RBS, repressing translation, enabling the culture to propagate. When an attempted thermal shift of 42°C triggers the instant unzipping of the stem loop structure (Kortmann and Narberhaus, 2012) and the unmasking of the RBS (Neupert *et al*., 2008), translation is activated and production is initiated.

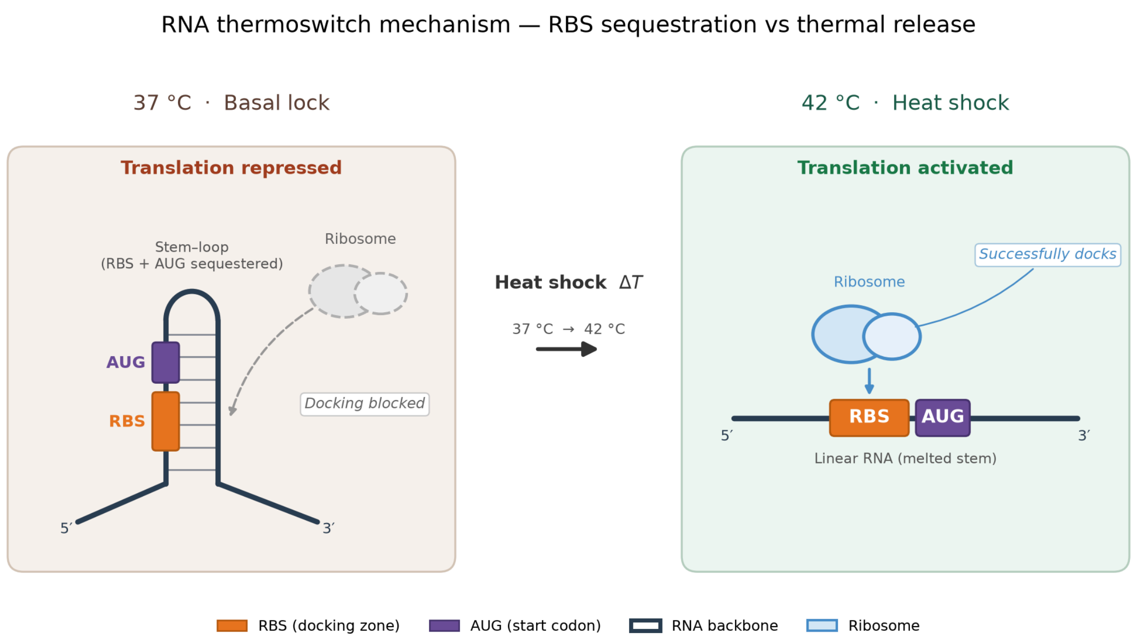

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Figure 1: RNA thermoswitch Mechanism: At 37°C, the RNA strand folds into a stable hairpin structure that sequesters the Ribosome Binding Site (RBS), creating a physical lock that

represses translation. Upon a 42°C heat shock, the stem-loop melts into a linear conformation, exposing the RBS for ribosomal docking and initiating chemical synthesis

1.2 Computational Bottleneck: Limitations of Predictive Classification 5

Initial efforts began with predicting RNATs using supervised Random forest (RF) classifiers. The RF were fed both established and well-known RNAT families from Rfam (Kalvari *et al*., 2021) (1,196 sequences) and RefSeq (1,196 negative control set) to predict viable thermoswitches using baseline thermodynamic metrics of a thermoswitch via NUPACK (Fornace *et al*., 2025) and ViennaRNA (Lorenz *et al*., 2011), which ended up being insufficient. Traditional classifiers rely on static scalar features such as minimum free energy(MFE) at a single temperature, which is not effective as 6 RNA is not dependent on a single rigid structure. It forms a dynamic, multi-state thermodynamic complex that changes conformation in response to thermal energy, meaning a static snapshot at 37°C simply cannot capture the continuous 37–42°C melting trajectory required for a heat-shock response (Neupert *et al*., 2008) Consequently, basic predictive models cannot calculate how an RNA structure behaves during a temperature shift. They cannot determine the exact temperature at which the 7 switch opens, nor how abruptly it triggers due to the fact that they rely on a single temperature snapshot rather than mapping the continuous melting curve. Thus static classifiers are fundamentally unequipped to engineer dynamic thermoswitches.

1.3 *De Novo* design and Validation

To overcome the limitations of static classification, synthetic bioengineering is increasingly pivoting towards deep generative design. Rather than filtering through established biological databases, generative models explore unconstrained sequence space. For this project, the Evolutionary Versatile Architect (EVA) was deployed (Huang *et al*., 2026). EVA utilises deep latent-space representations to sample and construct entirely novel *de novo* RNA structures. This approach effectively untethers the design process from the evolutionary biases present in natural datasets.

However, deep generative models often introduce a new problematic bottleneck: the validation gap. Architectures like EVA excel at generating topologically diverse sequences that possess baseline biological plausibility. Yet, they are fundamentally physics-blind (Huang *et al*., 2026). These models generate sequences based on learned statistical distributions, not dynamic multi-state thermodynamics. Generating a structure that merely looks like a hairpin is insufficient: geometric plausibility does not guarantee thermodynamic switchability (Abir and Zhang, 2025). An EVA-generated sequence might fold perfectly at 37°C, but completely fail to melt at 42°C. Because wet-lab synthesis of these untested candidates is prohibitively expensive and time-consuming, evaluating these generative outputs remains a major hurdle.

A computational bridge is therefore required. Generative outputs must be rigorously filtered through a dynamic, physics-based evaluation system before physical manufacturing is ever considered. By engineering an automated *in silico* biophysical pipeline, the continuous thermodynamic ensembles of generated candidates can be simulated across a strict temperature gradient. This allows for the extraction of Hill function metrics to mathematically verify switching steepness and activation thresholds. Ultimately, this pipeline translates the raw, structural outputs of a latent-space model into functional, dynamically verified bio-components.

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2. Data and Tools Used

2.1 Data

To evaluate supervised classification and benchmark *de novo* design, two distinct sequence cohorts were curated: an established positive RNAT set, a balanced negative control set

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Labelled Rfam / RefSeq panel for supervised RF training

Cohort	Source	Size (N)	Target Label	Feature Representation	Role in Workflow
Positive RNATs	Rfam	1,196	Positive (1)	Static Tabular Features	Supervised RF Training
Negative Controls	NCBI RefSeq	1,196	Negative (0)	Static Tabular Features	Supervised RF Training

Table 1: Dataset composition for the supervised Random Forest Model

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2.1.1 Positive Class: Rfam Database

The positive training dataset comprises 1,196 naturally occurring prokaryotic RNAT sequences extracted from the Rfam database (Kalvari *et al.*, 2021). Rfam organises non-coding RNA into functional families using Covariance Models (CMs) (Eddy and Durbin, 1994)—which extend profile Hidden Markov Models (Eddy, 2011)—and are searched and aligned using the Infernal software suite (Nawrocki and Eddy, 2013). These statistical models employ profile stochastic context-free grammars to capture both primary sequence conservation and secondary base-pairing dependencies. Sequences were harvested across well-characterised thermoswitch families, including the FourU, ROSE, and prfA thermometer motifs (Kortmann and Narberhaus, 2012), which naturally regulate heat-shock and virulence gene expression in bacterial pathogens (Loh *et al.*, 2018)."

2.1.2 Negative Class: RefSeq Background Controls

Supervised binary classification requires an equally representative negative
12 distribution. A negative cohort of 1,196 non-switching bacterial 5’ untranslated regions (5’ UTRs) and mRNA transcripts was sampled from the NCBI Reference Sequence
13 (RefSeq) database. This database serves as the gold standard for comprehensively annotated, non-redundant transcriptomic baselines. Naive negative selection introduces severe statistical artefacts. To eliminate trivial shortcut learning, RefSeq sequences were clustered at 80% identity using CD-HIT (Fu *et al.*, 2012), screened for sequence homology using BLAST (Altschul *et al.*, 1990) via the command-line suite (Camacho *et al.*, 2009), and filtered to match the positive class distribution in both sequence length (ranging between 60 and 180 nucleotides) and GC-content—addressing the known risk that raw MFE scales as a length proxy (Trotta, 2014). This ensures the classifier is forced to evaluate true thermodynamic folding characteristics rather than simple base composition biases.

2.2 Tools

2.2.1 Core Tools

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2.2.1.1 Python (v3.12.5)

The primary interpreter and runtime environment, with this specific version locked to ensure strict dependency resolution across the machine learning and bioinformatics pipelines.

2.2.1.2 Pandas (v3.0.5) & NumPy:

Primary tabular data structures for feature matrix construction, sequence deduplication, and multidimensional numerical arrays

2.2.1.3 SciPy

For advanced statistical operations and non-linear curve fitting, specifically for extracting the Hill equation metrics from continuous thermal data.

2.2.1.4 BioPython

Primary biological data input/output system, parsing FASTA files and managing sequence input/output (IO) for the raw datasets.

2.2.1.5 Conda CLI

A suite of command-line bioinformatics tools, including CD-HIT (Fu *et al.*, 2012), BLAST (Altschul *et al.*, 1990; Camacho *et al.*, 2009), HMMER (Eddy, 2011), and Infernal (Nawrocki and Eddy, 2013), was deployed outside the Python environment for sequence homology searching, alignment, and dataset clustering.

2.2.2 Random Forest Tools

2.2.2.1 Scikit-Learn (v1.9.0)

The core machine learning library utilised to construct, train, and validate the baseline RF classifier and run cross-validation

2.2.2.2 ViennaRNA & NUPACK

The core biophysical physics engines integrated into the classification pipeline to extract static MFE and ensemble thermodynamic features from the RNA sequences (Lorenz *et al*., 2011; Fornace *et al*., 2025).

2.2.3 EVA Cloud Deployment Tools

2.2.3.1 PyTorch & CUDA

Deep learning tensor framework for the Evolutionary Versatile Architect (EVA), enabling hardware-accelerated causal language modelling via GPU clustered virtual machines (Huang *et al.*, 2026).

2.2.3.2 Transformers & Hugging Face

A fast tokenisation backend and responsible for the direct loading of the pre-trained, multi-billion parameter EVA neural network weights from the GENTEL-Lab repository (Huang *et al.*, 2026).

2.2.3.3 Boto3 & Cloud Infrastructure

Main orchestrator tool for remote compute environments, managing secure credential loading via python-dotenv and facilitating large-scale artifact transfers to AWS S3 storage buckets

3. Methods and Tools

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3.1 Pipeline Architecture

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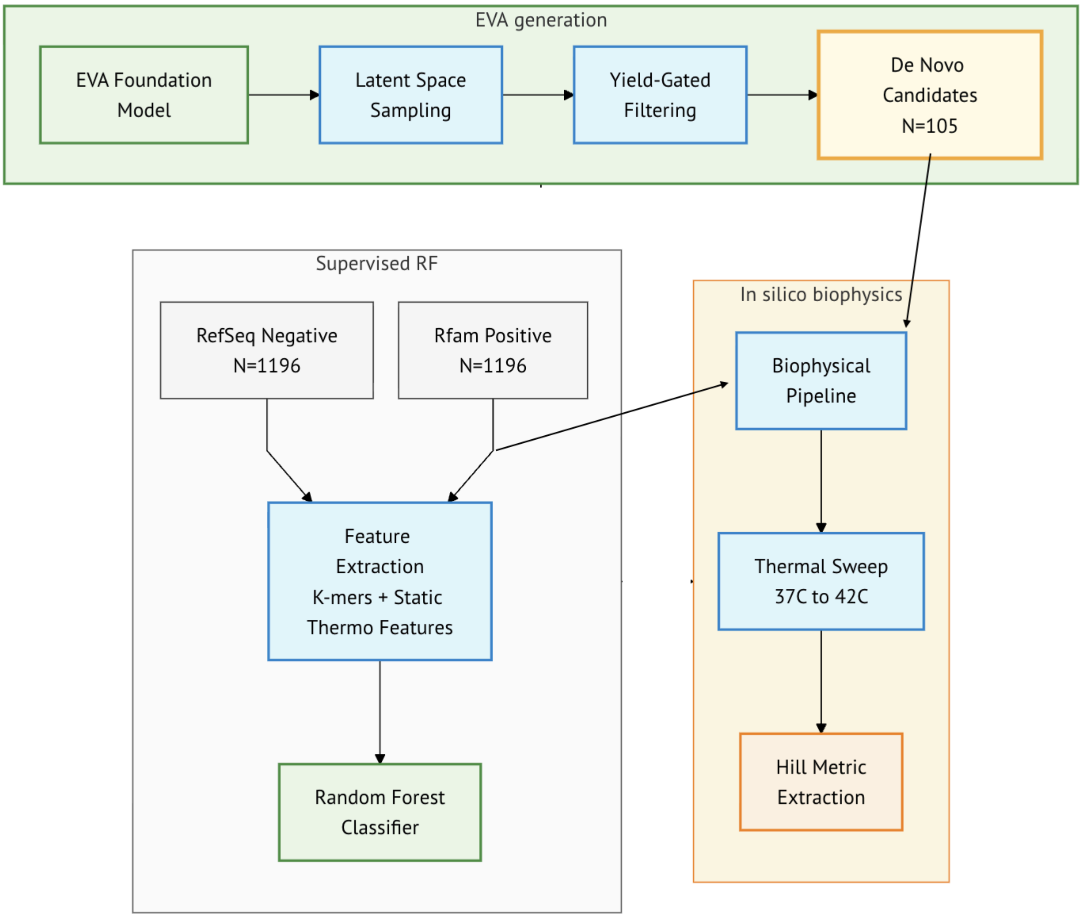

Figure 2: Flowchart of the computational architecture. The flowchart displays three grouped domains: EVA generation, Supervised RF, and *In silico* biophysics and the overall architecture of *de novo* and comparison metrics calculated mainly via NUPACK (Fornace *et al*., 2025) and ViennaRNA (Lorenz *et al*., 2011).

3.2 Supervised Predictive Baseline: Random Forest Classification

Utilizing the perfectly balanced, length- and GC-matched cohort established in Section 2.1, the sequences were transformed into a standardised numerical matrix to train the supervised classification baseline.

3.2.1 Sequence and Thermodynamic Feature Extraction

The core of evaluating RNA structural viability lies in translating a linear string of nucleotides into meaningful physical and evolutionary parameters. To achieve this, the sequences were parsed into two distinct feature domains, capturing both their localized grammar and their physical architecture.

K-mer Frequencies: First, to capture the localised "vocabulary" of the sequences, overlapping dinucleotide ($k = 2$) and trinucleotide ($k = 3$) frequencies were extracted for each transcript. These k-mers allow the classifier to recognise the evolutionary signatures and structural motifs heavily conserved across natural thermoswitch families. 17

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Static Thermodynamic Scalars: However, sequence grammar alone does not dictate physical folding. To map the baseline structural architecture, the sequences were evaluated at a static physiological temperature of 37°C utilising the ViennaRNA (Lorenz *et al.*, 2011) and NUPACK (Fornace *et al.*, 2025) software suites. Key extracted metrics included the ensemble free energy ($\Delta G_{ensemble}$) and the MFE(ΔG_{MFE}).

However, an unexpected complication emerged during the initial feature evaluation. While the intention was to map the physical architecture, it became apparent that the raw MFE (ΔG_{MFE}) harboured a hidden bias. Because longer RNA sequences naturally form more base pairs, their raw ΔG_{MFE} is mathematically driven to lower (more stable) negative values. It was realised that if fed directly into a classifier, the model would inevitably take the path of least resistance, exploiting this thermodynamic scalar not as a true measure of structural stability, but as a trivial proxy for sequence length. Uncovering this computational loophole prompted an immediate data engineering intervention. To completely neutralise the shortcut, the raw MFE was strictly length-normalised to extract the true thermodynamic density:

$$\Delta G_{density} = \frac{\Delta G_{MFE}}{N}$$

where N represents the total sequence length in nucleotides (Trotta, 2014). This crucial transformation forces the downstream algorithm to evaluate the genuine folding efficiency of the molecule, ensuring it learns actual biophysics rather than simply counting nucleotides.

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3.2.2 Random Forest Classification

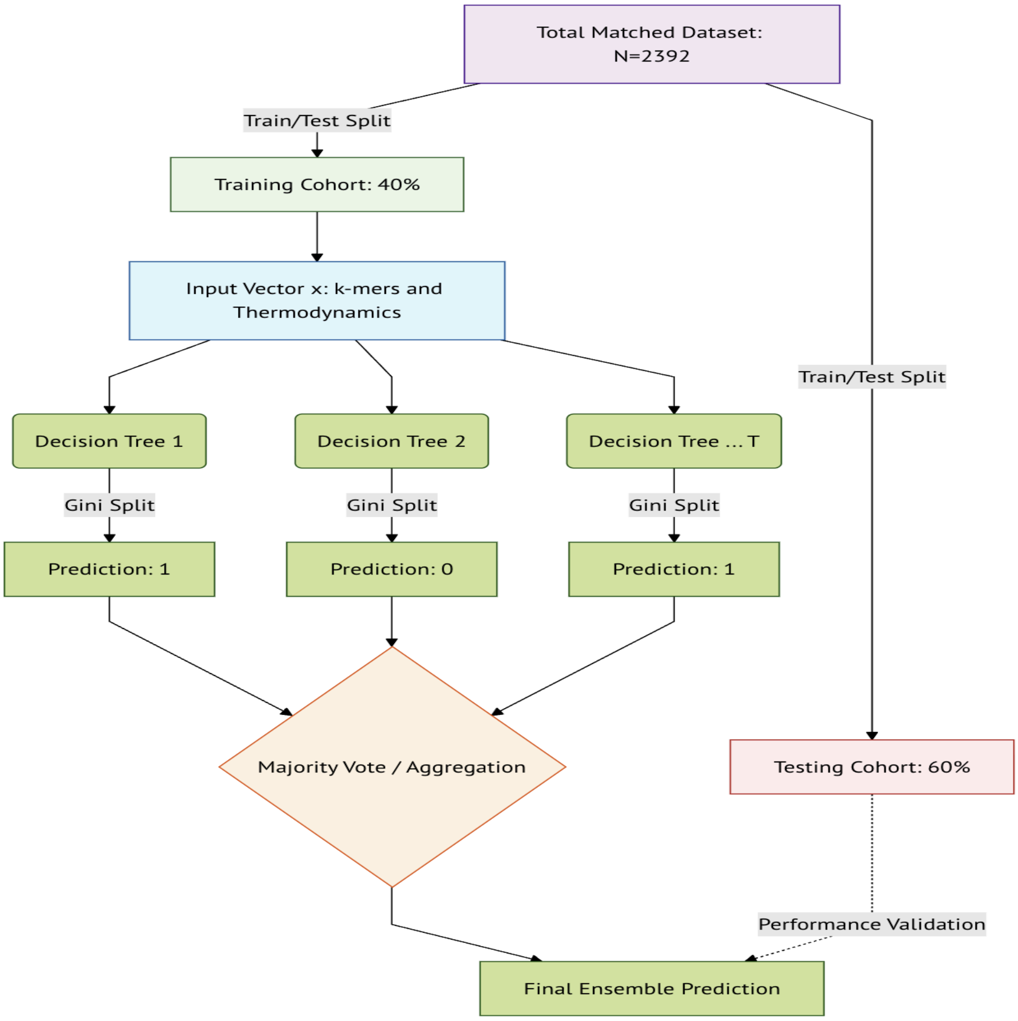

Figure 3: Flowchart of Random Forest architecture: The diagram outlines the 40/60 train-test split and the ensemble pathway, utilizing feature bagging with replacement across independent decision trees to prevent overfitting

To establish the baseline, a RF classifier was deployed via scikit-learn (Pedregosa *et al.*, 2011). The RF algorithm constructs an ensemble of decision trees utilizing bootstrap aggregating (bagging). For each tree, a random sample of the training data is drawn with replacement, ensuring that individual trees remain functionally decoupled and robust against variance within the dataset.

Furthermore, to decorrelate the trees, the model employs feature bagging. At each node split, only a random subset of the k-mer and thermodynamic features are evaluated. The optimal split is determined by minimizing the Gini impurity (G), which quantifies the probability of misclassifying a randomly chosen sequence if it were labelled according to the distribution of classes in that node:

$$G = 1 - \sum_{i=1}^{C} p_i^2$$

where C is the number of classes (for binary classification, $C = 2$) and p_i is the empirical fraction of items belonging to class i at the given node.

For a given RNA sequence represented by feature vector x, the final architecture outputs a probability P of belonging to the positive thermoswitch class based on the aggregated vote of T individual decision trees:

$$P(Thermoswitch|x) = \frac{1}{T}\sum_{t=1}^{T} h_t(x)$$

where $h_t(x) \in \{0, 1\}$ represents the binary prediction of the t-th tree. The final ensemble utilized $T = 200$ trees with unconstrained maximum depth, operating with a random state of 42. Model performance was rigorously evaluated using 5-fold stratified group cross-validation. Grouping was enforced at the RNA covariance family level (e.g., FourU, ROSE) (Kortmann and Narberhaus, 2012) to guarantee that homologous sequences remained isolated within individual folds. This explicitly prevented sequence homology data leakage between the training and validation sets, while stratification maintained precise class equilibrium. This ensured the model's predictive capacity was benchmarked on genuinely unseen structural architectures utilizing solely static, 37°C snapshot data.

3.3 Deep Generative Design: EVA *De Novo* design

The predictive baseline of the RF model demonstrated the inherent limitations of evaluating RNA thermoswitches through static, discrete feature partitioning. To achieve genuine structural innovation, the Evolutionary Versatile Architect (EVA) foundation model was deployed (Huang *et al*., 2026). Rather than classifying existing topologies, EVA functions as a deep generative framework, mapping the discrete combinatorial space of RNA nucleotides into a continuous, differentiable latent manifold.

3.3.1 Foundation Model Architecture and Cloud Infrastructure

EVA bypasses traditional biological sequence alignment by treating RNA generation as a causal language modelling task conditioned on RNA-type and taxonomic tags (Huang *et al*., 2026). Because evaluating and sampling from a multi-billion parameter neural network exceeds standard local compute limits, the model was deployed via a remote cloud architecture.

Pre-trained weights were provisioned to hardware-accelerated GPU virtual machines utilizing PyTorch and CUDA environments to enable high-throughput inference. Generation workflows and secure credential loading were orchestrated via Python and Boto3, facilitating the automated transfer of the generated *de novo* sequence artifacts to AWS S3 storage buckets for subsequent local downstream triage.

3.3.2 Foundation Model Architecture and Autoregressive Generation

EVA bypasses traditional biological sequence alignment by treating RNA generation as a causal language modelling task (Huang *et al*., 2026) conditioned on a continuous latent space. Pre-trained weights for the multi-billion parameter network were provisioned to hardware-accelerated virtual machines, enabling high-throughput inference. 20

By sampling a latent vector z from the continuous distribution, the model generates *de novo* RNA transcripts. The sequence generation is autoregressive, where the probability of predicting the next nucleotide x_i at position i is conditioned on both the latent vector and the preceding sequence context:

$$P(S) = \prod_{i=1}^{L} P(x_i | x_1, x_2, ..., x_{i-1}, z)$$ 21

This probabilistic sampling allows the framework to traverse regions of the evolutionary landscape that are completely unrepresented in natural sequence databases such as Rfam. (Kalvari *et al*., 2021).

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3.3.3 Yield-Gated Triage and Candidate Isolation

Unconstrained generation from a deep language model invariably produces a high volume of unviable or structurally unstable sequences (Huang et al., 2026). From an initial raw generative output of 2,000 accepted EVA sequences, a rigorous soft-drop protocol was applied to remove fatal structural anomalies. This included discarding transcripts with non-canonical compositions, length outliers outside the range of 40 to 600 nucleotides, and low-complexity collapses defined by a mono/dinucelotide fraction ≥ 0.85 or a unique 3-mer ratio < 0.05 23 24

To isolate functional and highly novel thermoswitch candidates, the remaining sequences were subjected to a strict yield-gated triage filter requiring a z-score Z ≤ -2, a positive change in unzipping probability ($\Delta P_{RBS} > 0$), and a maximum expectation value $E_{Rfam} > 10^{-3}$ against the biological database. Following this filtration, a final 25 cohort of $N = 105$ *de novo* thermoswitch candidates (a 5.25% retention rate) was isolated for direct benchmarking against the biological ground truth. High *in silico* attrition of this scale is standard in generative RNA design: Joshi *et al*. (2025) generated approximately one million sequences and computationally filtered them down to 1,000 candidates. Structural novelty against Rfam (Kalvari *et al*., 2021) was evaluated using nhmmer (Wheeler and Eddy, 2013), aligning with the profile homology screening protocols established by Zhao *et al*. (2024).

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ViennaRNA (Lorenz *et al.*, 2011) was selected as the primary thermodynamic gate for generative triage; out-of-band analysis (Appendix A) confirmed that while deterministic engines agree on static structural MFE, they fundamentally lack cross-engine consensus for dynamic melting phenotypes, necessitating a single-engine primary filter.

3.4 *In Silico* Biophysical Validation and Hill Function Modelling

To overcome the static limitations of the predictive baseline and rigorously evaluate the 105 *de novo* sequences generated by the EVA model, a dynamic *in silico* validation pipeline was engineered. This phase maps the continuous thermodynamic trajectory of each candidate, simulating the precise biological heat-shock window required for industrial biomanufacturing.

3.4.1 Dynamic Thermal Sweep Simulation

Unlike standard classifiers that evaluate a single folding state, functional thermoswitches rely on multi-state structural ensembles. To simulate this, the candidates and the natural Rfam controls were subjected to a continuous thermal

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sweep from T = 37°C to T = 42°C (Neupert *et al*., 2008) utilizing the ViennaRNA (Lorenz *et al*., 2011) and NUPACK (Fornace *et al*., 2025) partition function algorithms.

At each temperature step T, the thermodynamic ensemble is computed to determine the probability that the Ribosome Binding Site (RBS) remains structurally unpaired (and

thus accessible for translation). The probability $P_{unpaired}$ at a given temperature is derived from the statistical mechanical partition function Z:

$$P_{unpaired}(T) = \frac{Z_{unpaired}(T)}{Z_{total}(T)}$$

where Z_{total} represents the sum of the Boltzmann weights for all possible secondary structures, and $Z_{unpaired}$ is the restricted partition function for states where the critical RBS nucleotides remain unbound. This process generates a continuous melting trajectory for every sequence.

3.4.2 Hill Function Modelling and Metric Extraction

To quantitatively benchmark the performance of the generated sequences against the natural controls, the raw thermodynamic melting trajectories must be mathematically parameterized. The transition of an RNA thermoswitch from a locked state to an open state exhibits cooperative melting behaviour (Kortmann and Narberhaus, 2012), which is optimally modelled using an adapted Hill equation.

Using non-linear least squares optimization, each sequence's thermal trajectory was fitted to the following logistic Hill function:

$$y(T) = y_{min} + \frac{y_{max} - y_{min}}{1 + (\frac{T_m}{T})^h}$$

where:

- $y(T)$ is the probability of the RBS being unpaired at temperature T.
- y_{min} and y_{max} represent the baseline structural leakiness (basal state at 37°C) and the maximum unzipping capacity, respectively.
- T_m (Activation Threshold) is the critical melting temperature at which the switch achieves 50% of its maximum conformational change.

- h (Hill Coefficient) quantifies the cooperativity and steepness of the structural transition.

To ensure computational stability across highly variable thermodynamic ensembles, the fit was executed using scipy.optimize.curve_fit. Because data-driven box bounds were established—specifically capping $T_m \in [5, 95]$ and $h \in [0.1, 20]$—the algorithm defaulted to the Trust Region Reflective (TRF) optimization method, capped at 10,000 maximum function evaluations.

3.4.3 Validation Criteria

By extracting h and T_m, the pipeline translates raw structural outputs into comparable mathematical metrics. A viable *de novo* thermoswitch must not only exhibit a heat-inducible T_m strictly aligned with the 42°C heat shock trigger (Neupert *et al.*, 2008), but also possess a high Hill coefficient (h), indicating a sharp, instantaneous phase transition rather than a gradual, leaky unravelling. It must be noted that while heat-repressible RNase E designs (Hoynes-O'Connor *et al.*, 2015) provide a useful template for how RNA structures break and reform in response to temperature, they are engineered to turn *off* when heated. This makes their physical unzipping mechanics the exact opposite of the heat-triggered "ON" switches targeted by the EVA model. Ultimately, these extracted metrics form the definitive comparative benchmark for Chapter 4, detailing exactly where the foundation model succeeded in topology, but failed the rigorous demands of dynamic biophysics.

4. Results

4.1 Random Forest results

4.1.1 Cohort Matching and Control Baselines

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Model	Features	Stratified ROC-AUC	Accuracy	vs chance (0.50)
Length-alone RF	seq_length	0.196	0.273	Below chance
%GC-alone RF	%GC	0.254	0.279	Below chance
Length + %GC RF	length + %GC	0.157	0.234	Below chance
Random chance (balanced)	–	0.500	0.500	–

Table 2 Baseline compositional ablation and control evaluation: The table detailing the predictive performance of RF models trained different isolated compositional features

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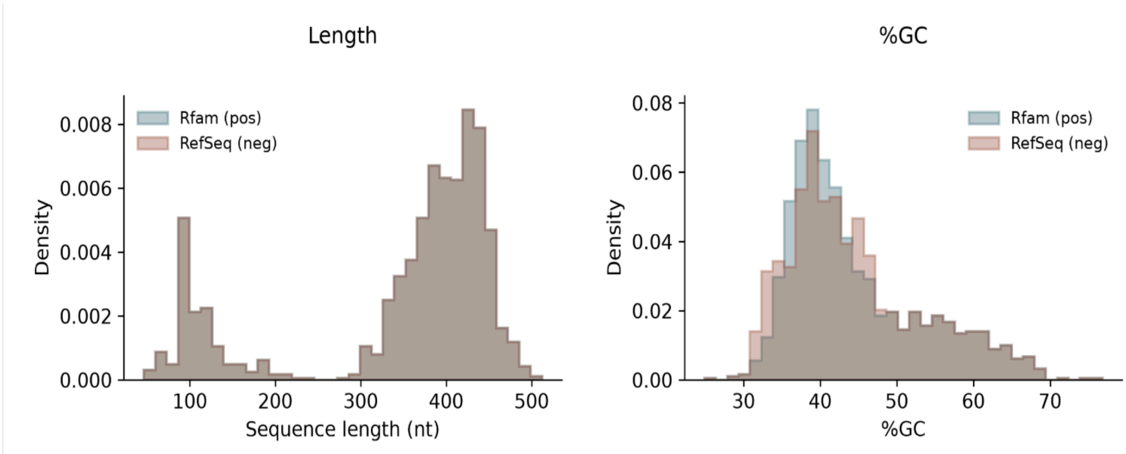

Figure 4 Overlaid probability density histograms of matched sequence cohorts: distributional overlap of sequence length (nt) and percentage GC content (%GC) between the Rfam positive thermoswitches and the RefSeq negative controls

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Prior to evaluating the predictive baseline, it is critical to verify that the negative background cohort does not introduce trivial classification shortcuts. Table 2 demonstrates the efficacy of the Hungarian matching protocol (Kuhn, 1955). When the RF was trained exclusively on sequence length, it yielded a stratified ROC-AUC of 0.196 (Fawcett, 2006), mathematically proving that sequence length cannot separate the classes. 31 Furthermore, the overlaid density distributions confirm that the RefSeq negative controls physically mimic the Rfam positive class across both sequence length and percentage GC content. This strict compositional matching forces the downstream classifier to evaluate underlying structural features rather than basic compositional biases.

4.1.2 Supervised Baseline: Cross-Validation and Group Holdout Collapse

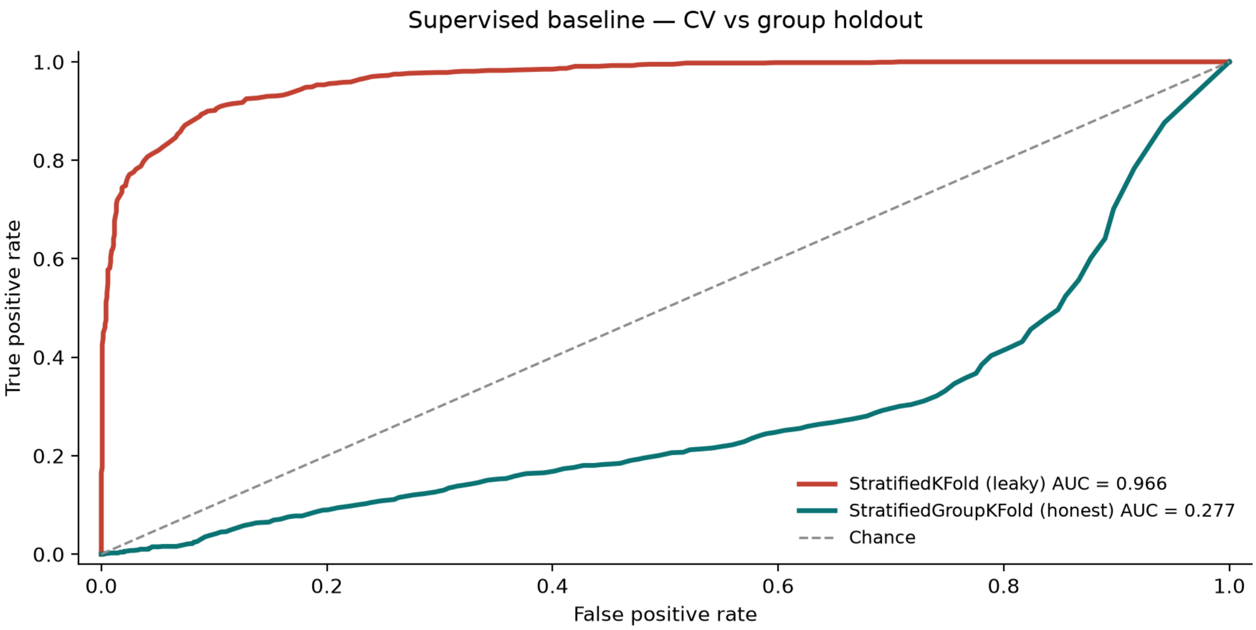

Figure 5 Receiver Operating Characteristic (ROC) curves evaluating cross-validation protocols: This is detailing the collapse of the predictive performance by showcasing the contrast between standard StratifiedKFold protocol (AUC = 0.966) against the homology-isolated StratifiedGroupKFold (AUC = 0.277)

With trivial shortcuts eliminated, the RF model was evaluated using a 92-feature non-circular matrix. When evaluated under standard stratified cross-validation, the model achieved an apparent ROC-AUC of 0.966 (Fawcett, 2006). However, this metric initially, was highly misleading. Because natural RNA thermoswitches are heavily conserved, standard cross-validation allows the model to memorize sequence homologies shared across the training and testing splits. When honesty was enforced via StratifiedGroupKFold—grouping by the RNA family to completely isolate homologous sequences—the ROC-AUC collapsed to 0.277. This visually stark drop-off
32 proves that the model failed to learn a transferable, global rule for thermoswitch physics, and instead heavily overfit to the training families.

4.1.3 Feature Attribution and the K-mer Sequence Memorisation

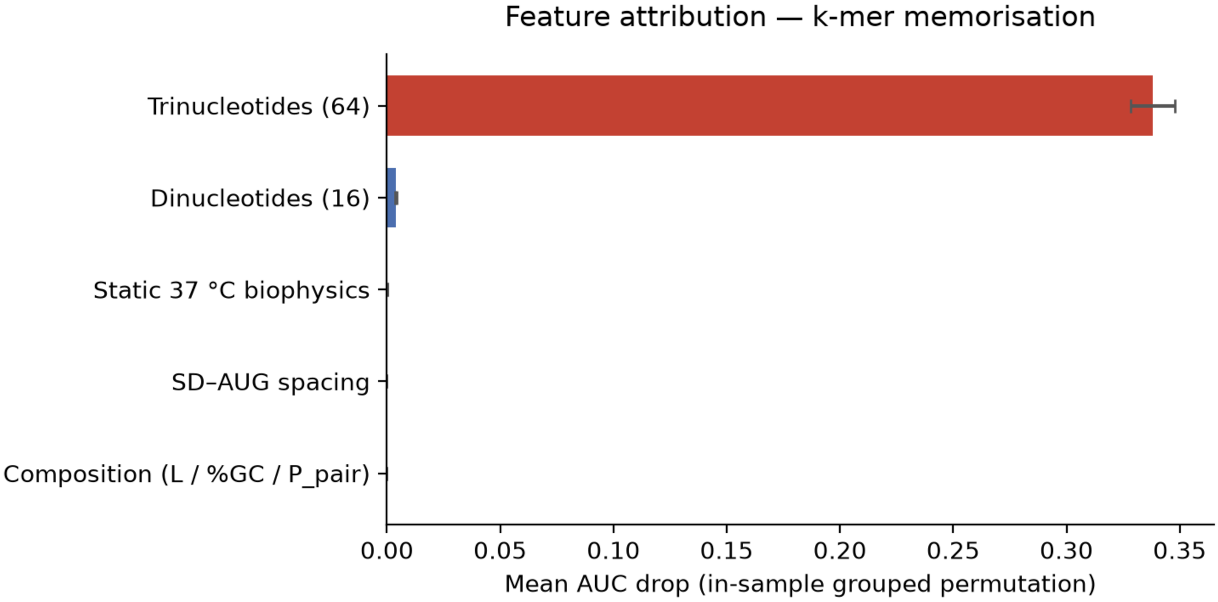

Figure 6 Grouped permutation importance of the Random Forest classifier: Mean ROC-AUC drop occurred when specific features were shuffled and permuted out revealing the dominance of the trinucleotide feature set and the little to no effect of the other features including Static 37°C biophysics which incl. ΔGMFE/N.

To understand here the biological explanation for this cross-validation collapse, grouped permutation importance was extracted from the full-fit forest. The attribution chart reveals that the model's discriminative power was almost entirely localized 33
within the 64-dimensional trinucleotide frequency block, which drove a mean AUC drop of 0.338. In contrast, the static 37°C biophysics and sequence composition blocks contributed negligibly, with AUC drops of approximately zero. This confirms the memorisation trap: the static classifier bypassed thermodynamic rules entirely and 34
relied on family-correlated trinucleotide motifs to separate the classes.

In the Methodology, rigorous biophysical constraints were applied—such as length-matching the cohorts and normalizing the free energy to thermodynamic density ($\Delta G_{MFE}/N$) (Trotta, 2014)—to force the RF to learn genuine RNA physics. However, the feature attribution reveals the inherent greediness of the algorithm. Despite perfectly isolating the thermodynamic signals, the model completely ignored the physics. The evolutionary signatures within the K-mers provided such a pristine 35 data leak that the model bypassed the physical architecture entirely, opting instead for trivial sequence memorisation.

4.1.4 Confidence Calibration and Out-of-Distribution Transfer

The operational consequence of this memorisation is extreme model uncertainty when faced with out-of-distribution data. An analysis of the out-of-fold predicted probabilities ($\hat{y}$) from the honest StratifiedGroupKFold protocol shows a massive concentration of scores in the low-confidence bin, representing 1,504 sequences. Conversely, the high-confidence bin ($\hat{y} \geq 0.80$) contained only 6 sequences. Consequently, while the RF can serve as a soft ranking aid, it completely lacks the calibration required to act as a definitive wet-lab discovery gate. 30

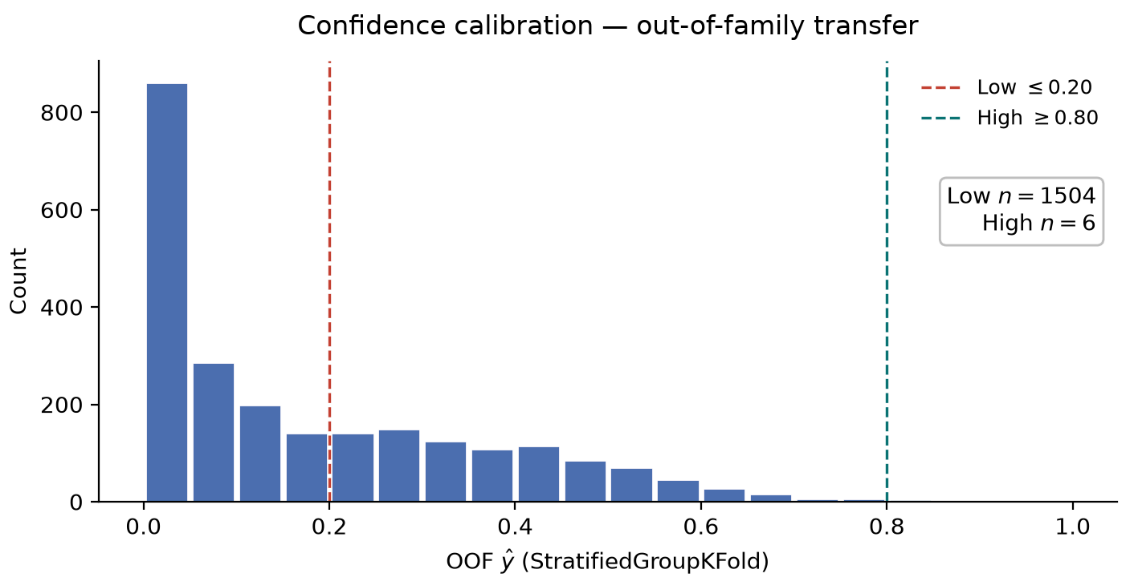

Figure 7 Out-of-fold prediction histogram of the RF model: Majority of the out of family sequences where the feature k-mers is silenced, resulting in predicting sequences that are not thermoswitches at near zero (≥0.2) 37

4.2 EVA results

often ?

The failure of the discriminative classifier demonstrated that static models inherently default to trivial sequence memorisation. To successfully engineer novel RNA thermoswitches, the framework must instead internalise the underlying structural and 38 evolutionary grammar. To achieve this, the deep generative EVA framework was deployed. However, generative sequence sampling necessitates rigorous post-hoc validation to separate functional biophysics from statistical hallucinations.

4.2.3 Generative Viability: EVA Generative Triage

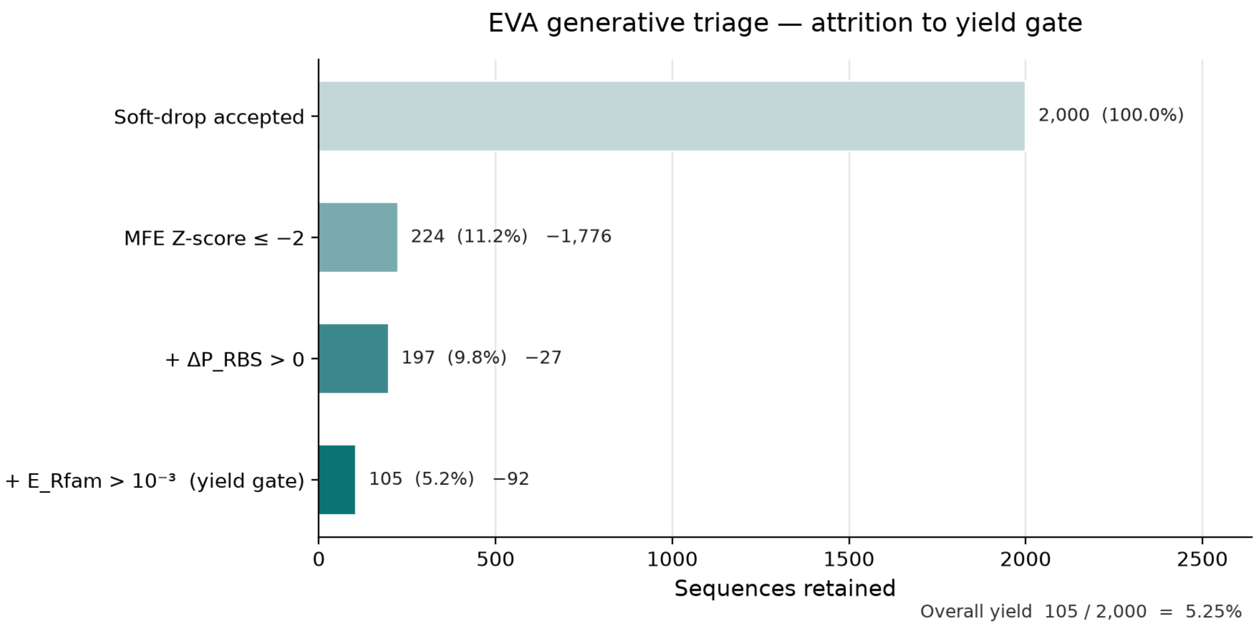

Figure 8 generative triage and biophysical attrition funnel: A filtering schematic that explains the attributable difference between genuine plausible thermoswitch-like structures from hallucinations. Candidates were first filtered for baseline structural stability (MFE Z-score ≤ -2) 39), followed by the requirement for dynamic conformational shifting ($\Delta P_{RBS} > 0$), and finally gated for evolutionary viability ($E_{Rfam} > 10^{-3}$).

To bypass the limitations of classification, the deep generative EVA model produced a raw cohort of 2,000 soft-drop accepted sequences. However, because deep language models are fundamentally physics-blind (Huang *et al.*, 2026), these topologically plausible outputs required stringent filtering. The automated triage pipeline applied a strict yield gate requiring a favorable MFE (Z ≤ -2), a positive opening probability ($\Delta P_{RBS} > 0$), and a novelty check against the Rfam database ($E_{Rfam} > 10^{-3}$). As visualized in the attrition funnel, this rigorous evaluation filtered out the vast majority of candidates, successfully isolating a highly refined cohort of 105 *de novo* thermoswitches (a 5.25% overall yield) for dynamic validation. A ~95% kill rate is therefore not a failed generator but a recognised generative-filtering step (Joshi *et al.*, 2025); the Rfam novelty arm of the same gate follows the nhmmer homology protocol used for GenerRNA (Zhao *et al.*, 2024; Wheeler and Eddy, 2013).

4.3 Thermal Validation: *In Silico* Thermal Sweeps

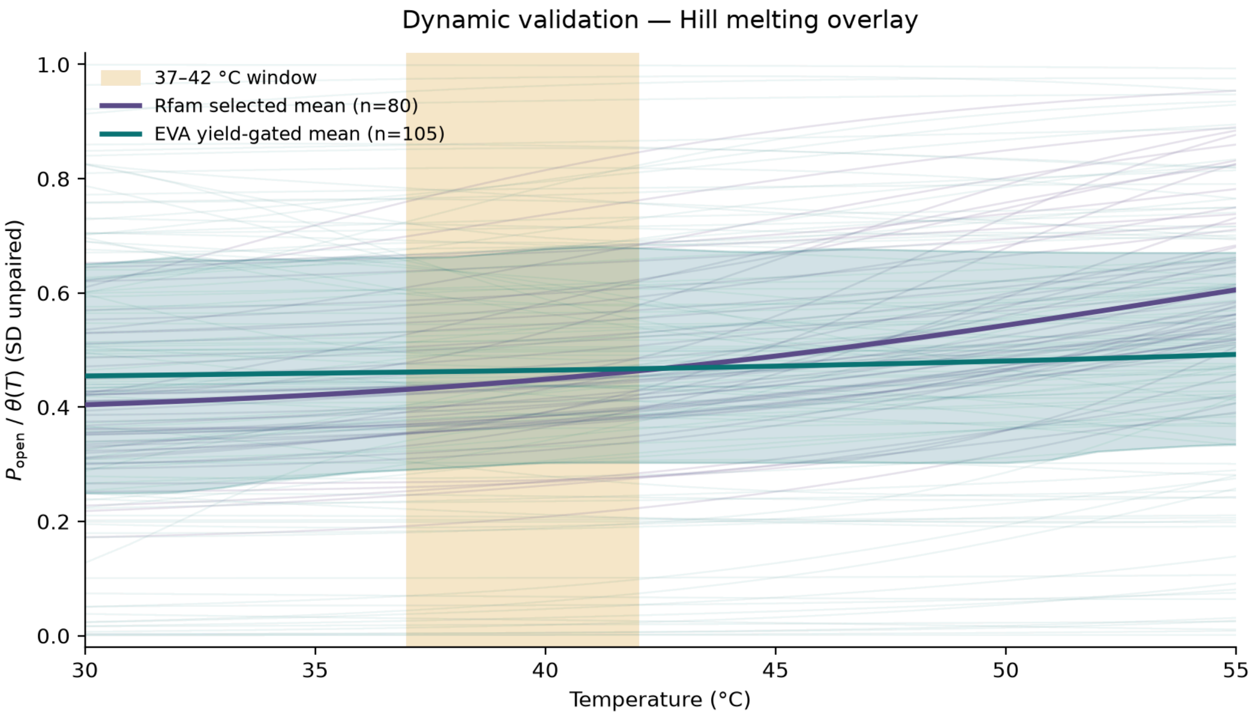

Figure 9 Dynamic thermal validation and Hill melting overlay: The solid lines represent the respective cohort mean, while the shaded regions indicate the standard deviation (SD) of the unpaired probability (P_{open}) across the sequences at each temperature step. P_{open} quantifies the thermodynamic likelihood of the Ribosome Binding Site (RBS) remaining free from secondary structure blockade, representing the functional availability of the transcript for translation. (4)

The ultimate benchmark of functionality is the continuous thermodynamic melting trajectory. Subjecting the 105 yield-gated EVA candidates and the natural Rfam controls to a thermal sweep from 37°C to 42°C reveals the definitive validation gap. While the Rfam controls (purple) exhibit a rising, cooperative transition curve characteristic of biological thermoswitches (Kortmann and Narberhaus, 2012) , the EVA-generated candidates (teal) trace a consistently flat, unresponsive trajectory across the critical heat-shock window. This overlay confirms that while the foundation model excels at generating statically sound RNA topologies, it universally fails to capture the dynamic, multi-state biophysics required to trigger chemical synthesis at exactly 42°C (Neupert *et al*., 2008). In other words, EVA produced geometrically plausible hairpins that were not thermodynamically switchable in the industrial heat-shock window (Abir and Zhang, 2025; Huang *et al*., 2026).

5. Discussion

Our current pipeline relies on thermodynamic engines that evaluate RNA at a static, global equilibrium. However, in a real cell, RNA folding is a time-dependent process; the molecule folds dynamically as it is being transcribed (Twittenhoff *et al*., 2020). Since our static equilibrium engines (Lorenz *et al*., 2011) could not capture this kinetic reality, the model engineered sequences that were far too stable. A practical next step is to swap out these static thermodynamic engines for kinetic folding simulators. If the evaluation pipeline can track how a sequence folds over time, the algorithm could learn to design the temporary, spring-loaded structures required for a functional thermal switch.

Additionally, the current framework only evaluates 2D secondary structures, which essentially act as flat blueprints of base pairs. But real RNA melting relies heavily on 3D tertiary interactions—complex spatial tangles and loops—to execute a rapid phase transition. 42 Energy-aware 3D generators such as RNA-EFM were built precisely so that designed RNAs are thermodynamically stable rather than merely geometrically plausible (Abir and Zhang, 2025)—the same failure mode observed here. Since 2D models physically cannot represent these mechanics, future iterations of this pipeline should plug into emerging 3D prediction architectures, such as RoseTTAFold2NA or machine-learning-accelerated molecular dynamics. Upgrading the evaluation step to a full 3D environment would give the generative model the real-world spatial feedback it needs to design sequences that actually unzip.

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Finally, while generative models like EVA are excellent at mimicking the baseline patterns of their training data, they are not naturally built to engineer synthetic upgrades that fall outside of that data. Since our ultimate goal is to build a sharp, instant logic gate rather than a gradual biological rheostat, we need a more active optimization strategy. A highly viable next step is framing this as a Reinforcement Learning (RL) problem. As noted by Tang *et al.* (2026) and Runge *et al*. (2024), blind reinforcement learning often underperforms classical RNA-design methods unless biophysical constraints are built into the reward; therefore the RL agent must be constrained by a highly specific custom reward that scores the exact 37–42°C heat-shock window rather than a generic fold objective. Huang *et al*. (2026) further show that unconstrained EVA sampling from an unstructured global distribution is weak unless RNA-type and species tags suppress invalid draws—conditioning already used here, yet residual thermal inertia remains an EVA-era bottleneck.

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